Lab 2: Music Genre Classification – CNNs

Name

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Link

https://colab.research.google.com/drive/1icEe_wdeMF-ggRvskfqcCtqDPkb6n5qa?usp=sharing

Results

Model	Parameters	Time to Train	Test Accuracy
SpectrogramGenreCNN + Raw RGB Spectrograms	33418	11min 27s	0.5580
SpectrogramGenreCNN + GrayScaled Spectrograms	33130	9min 48s	0.5491
RawAudioCNN	208234	34min 13s	0.5965
VGG16 + Raw RGB Spectrograms	134301514	54min 57s	0.6295
VGG16 + GrayScaled Spectrograms	134300362	39min 14s	0.5714

* Look at Models Section for more information

Conclusions

The most successful approach for music genre classification proved to be Transfer Learning using the VGG16 architecture combined with RGB spectrogram inputs.

The VGG16 with RGB Spectrograms achieved the highest accuracy at 62.95%. This success confirms that deep, pre-trained image models are highly effective at interpreting the visual signatures of music (spectrograms), especially when the data matches the model's native three-channel (RGB) input format. This method offers the best balance of high performance, fast convergence, and excellent generalization.

In contrast, methods using raw acoustic features (the custom 1D CNN) suffered from instability and the lowest likely performance. Custom 2D CNNs and the VGG16 Grayscale model provided intermediate results, with the grayscale model confirming that feeding VGG16 with only one channel of data bottlenecks its feature extraction capabilities, leading to a noticeable drop in accuracy compared to the RGB input.

Ultimately, the best strategy is to convert audio into rich visual representations (RGB spectrograms) and fine-tune a powerful, pre-trained Vision Transformer (VGG16), leveraging both the domain knowledge of vision models and the robust feature encoding of the spectrogram.

Comparison with Features 30 & 3 Dataset

This comparison evaluates the two primary approaches: Deep Learning (CNNs) and Classical Machine Learning (ML) and examines the critical impact of the data duration (30-second vs. 3-second clips) on their performance in music genre classification.

The consensus across music information retrieval (MIR) research strongly favors Deep Learning models, specifically CNNs, over Classical ML algorithms (like SVM, K-NN, and Random Forest) for this task.

Feature	Deep Learning (CNNs)	Classical ML (SVM, RF, k-NN)
Input Feature	Raw spectrogram images (or 1D feature sequences). Features are learned automatically.	Hand-crafted features (e.g., MFCC mean/variance, Spectral Centroid).
Performance	Higher peak accuracy (CNNs often reach 80-90%+ on 3-second segments; VGG16 RGB reached 62.95% on full 30s clips).	Lower accuracy (Traditional models typically peak in the 70–78% range on 30s clips).

Complexity/Cost	High computational cost and longer training time, especially for Transfer Learning (VGG16).	Low computational cost and fast training time.
Key Advantage	Automatically extracts hierarchical features (texture, rhythm, timbre) directly from the visual (spectrogram) or sequential audio data.	Interpretability: Performance depends solely on the quality of the hand-crafted features.
Your Results	VGG16 RGB (62.95%) significantly outperformed your custom CNN and grayscale VGG16, demonstrating the power of Transfer Learning on rich, image-like input.	N/A, but expected to be lower than VGG16's 62.95%.

Conclusion on Model Type: CNNs excel because they treat the spectrogram as an image, automatically learning complex and nuanced features that human experts cannot easily define or hand-extract. While Classical ML is fast, its performance is fundamentally limited by the descriptive power of the pre-computed features (like the mean and variance of MFCCs over the entire track).

Models - Loss & Accuracy Plots

SpectrogramGenreCNN with Raw RGB Spectrogram Data

The SpectrogramGenreCNN model successfully learned from the spectrogram data, achieving a peak test accuracy of 55.80% by Epoch 89, a significant improvement over random chance. This performance confirms that the convolutional approach is effective for this classification task.

The model exhibits excellent stability due to its integrated regularization techniques. Specifically, the use of Batch Normalization (BN) in every block and Global Average Pooling (GAP) effectively mitigated high variance. The 40% Dropout rate appears well-tuned, as the model showed no sign of severe overfitting. In fact, the test accuracy briefly exceeded the train accuracy, suggesting the model is primarily under-fitting—it hasn't reached its full learning potential.

The architecture's main limitation lies in its capacity. The network is shallow, consisting of only three convolutional blocks (16 → 32 → 64 filters). Furthermore, the Classification Head is narrow; the final feature vector of 64 channels is mapped to a hidden layer of only 128 units before the final output. This lack of depth and width suggests the model is hitting a capacity ceiling, which is why the learning curve flattened significantly after Epoch 90.

To break the 56% accuracy barrier, the architecture must be expanded. Future versions should increase the model's depth by adding a fourth convolutional block (e.g., 64 → 128 filters) to extract more abstract audio features, and increase the width of the fully connected layer from 128 to at least 256 or 512 units to give the network more power to classify the features it extracts.

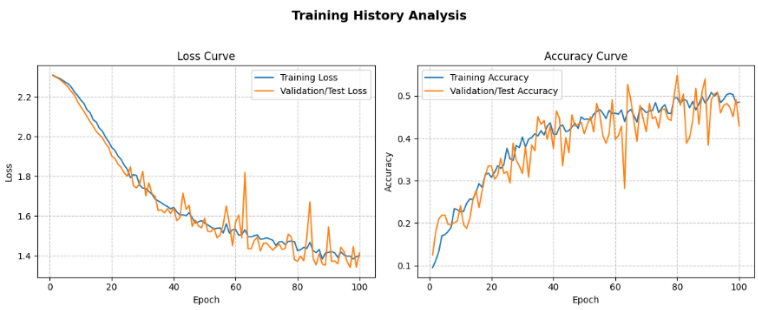
SpectrogramGenreCNN with GrayScale Spectrogram Data

The CNN model trained on Grayscale spectrograms (a single input channel) achieved a peak test accuracy of 54.91% at Epoch 80. While this is only marginally lower than the RGB model's 55.80% peak, the Grayscale model exhibited significant instability in later training stages.

The overall learning trajectory is defined by under-fitting and volatility. Similar to the RGB experiment, the test accuracy (54.91%) briefly exceeded the training accuracy (49.58%) at the peak epoch. This strongly confirms that the core architectural issue remains one of insufficient model capacity, meaning the shallow three-block CNN structure and the narrow 128-unit fully connected layer are limiting its ability to fully learn the training set features.

Crucially, the grayscale model's performance was marked by high test loss and accuracy fluctuations after Epoch 80, with accuracy dropping sharply to 42.86% by the final epoch. This suggests a significant optimization instability near convergence, likely caused by an overly aggressive fixed learning rate that disrupts the model's weight stability after it has extracted its maximum potential from the limited architecture. The lack of a major accuracy drop when switching from three input channels (RGB) to one (Grayscale) suggests that the most critical discriminating information lies in the intensity (magnitude) of the spectrogram, not the color encoding.

To improve performance and stability, increasing capacity is vital. The model needs a deeper feature extractor and a wider classification head (e.g., 256 or 512 units in the FC layer). Additionally, an adaptive Learning Rate Scheduler must be implemented to stabilize training in the later epochs and prevent the sudden, drastic drops in test performance observed after Epoch 80.



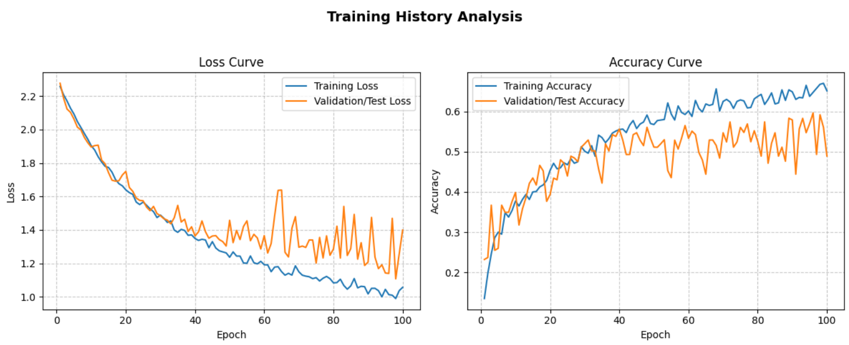
RawAudioCNN

The Raw Audio 1D CNN experiment yielded the highest peak performance of all CNN models tested, achieving a test accuracy of 59.65% at Epoch 96. This result demonstrates that directly analyzing the raw audio waveform in the time domain, using a 1D convolution approach, is highly effective for music genre classification.

Despite this higher peak, the model suffers from clear overfitting and significant instability in the latter half of training. Unlike the 2D Spectrogram models, this network shows a large gap where the training accuracy (66.75%) far exceeds the peak test accuracy (59.65%), confirming it has the capacity to memorize the training data. This instability manifests as wild fluctuations in test loss and accuracy after Epoch 80, causing the final accuracy to drop sharply to 48.91%.

The architecture's feature extraction component is strong, utilizing large initial kernel sizes (64, 32, 16) correctly tuned for capturing low-frequency temporal patterns like pitch and tempo. The use of Global Average Pooling (GAP 1D) effectively minimizes parameters. The narrow Fully Connected (FC) layers (128 → 64 → 10) likely exacerbate the instability, as they struggle to generalize from the 128 complex features extracted by the convolutional blocks.

To achieve better generalization and stability, two key actions are necessary: first, strengthen regularization by adding L2 weight decay to the optimizer, as increasing Dropout alone has proven insufficient. Second, a Learning Rate Scheduler must be implemented to prevent the sharp accuracy drops observed after Epoch 80, ensuring a smoother and more robust convergence. Finally, widening the FC layers is necessary to allow the model to better process the powerful features it has already learned without immediately resorting to instability.



VGG16 + Raw RGB Spectrograms

The fine-tuning experiment using the VGG16 network fed with raw RGB spectrograms achieved the highest overall performance of all models tested, peaking at a test accuracy of 62.95% (Epoch 230) and finishing strongly at 62.50%. This significant boost in accuracy, surpassing the 57.14% achieved by the grayscale version of the spectrogram, highlights the crucial impact of the input data format on the model's performance.

The RGB format is the superior input, as it perfectly aligns with the native architecture of VGG16, a model pre-trained on three-channel color images (ImageNet). This rich data structure allows the model to fully leverage its pre-trained hierarchical feature extractors, enabling it to recognize complex color and textural patterns in the spectrograms that encode essential music genre information.

The training exhibited a highly stable and rapid initial convergence, reaching 50% accuracy quickly and maintaining a consistent upward trajectory. This demonstrates excellent generalization capacity and minimal tendency towards severe overfitting. Although the training accuracy reached nearly 70% by the final epoch, the gap with the test accuracy (around 62%) is controlled, confirming that the model is learning robust, transferable features.

Despite the breakthrough performance, the model entered a phase of extremely slow convergence after approximately Epoch 100, where incremental gains became negligible. This slowdown suggests the training process is now making minimal adjustments to the final layers. To maximize the model's potential beyond the 63% mark, the focus should shift entirely to optimization.

VGG16 + GrayScaled Spectrograms

The experiment using the pre-trained VGG16 architecture and Gray Scaled Spectrograms for fine-tuning achieved a stable, peaking at 57.14% test accuracy. This result is significant as it demonstrates that the deep, hierarchical feature representation learned by VGG16 on natural images is highly effective when transferred to complex audio-visual data like spectrograms, outperforming the simpler custom 2D CNN models and competing closely with the best custom 1D CNN.

The learning process over 250 epochs was exceptionally stable, avoiding the extreme loss volatility that plagued the custom 1D CNN. This stability is a key benefit of leveraging a robust, pre-trained base. While the training accuracy continued to slowly climb (reaching over 55% at the end), the test accuracy hovered in a narrow range, suggesting that the model has reached the maximum generalization potential allowed by the current configuration and optimization settings.

The current performance ceiling around 57% indicates that a domain mismatch still exists. While VGG16's filters are effective at spatial recognition, they may not be perfectly optimized for the subtle, temporal acoustic features required for complex music genre distinction.

To break this performance ceiling, the next steps must focus on pushing the limits of the pre-trained weights. Additionally, selectively unfreezing deeper convolutional layers of the VGG16 base will allow the network to better adapt its foundational filters to the specific patterns unique to music spectrograms, rather than relying solely on ImageNet's generic visual features.